INVERTIS UNIVERSITY BAREILLY

Established by Govt. of U.P. u/s 2F of UGC Act, 1956 vide U.P. Act 22 of 2010.

STUDY AND EVALUATION SCHEME
(With effective from academic session 2023-2024)
BTech. in Cloud Computing
YEAR III, SEMESTER V

Sl. No.	Category	Course Code	Course Title/ Subjects	Hours per week			Evaluation Scheme		Total	Credits
				L	T	P	CA	EE		
THEORY										
1	Professional Core	BCSI501	Theory of Computation	3	1	0	30	70	100	4
2	Professional Core	BCSICT501	Cloud Computing	3	0	0	25	50	75	3
3	Professional Core	BCSICT502	Network Security	3	0	0	25	50	75	3
4	Professional Core	BCSICT503	Principles of Virtualization	1	0	0	10	15	25	1
5	HSM	BCSICT504	Humanities II	3	1	0	30	70	100	4
6	Professional Elective		Elective-I	3	0	0	25	50	75	3
7	Engineering Science Course	IIOT5	Deep Learning	4	0	0	30	70	100	4
PRACTICALS AND PROJECTS										
7	Professional Core	BCSICT508	Network Security Lab	0	0	2	10	15	25	1
8	Professional Core	BCSICT509	Principle of Virtualization Lab	0	0	2	20	30	50	2
9	Summer Training	BCSI502	Summer Project Seminar-II	0	0	2	20	30	50	2
			TOTAL	20	2	6	225	450	625	27

L-Lecture, **T**- Tutorial , **P**- Practical ,**CA**- Continuous Assessment, **EE**- End Semester Examination

Cloud Computing

L*-Lecture* , T*-Tutorial, P*-Practical

COURSE CODE	Elective - I
BCSICT505	Security Architecture
BCSICT506	Database Security

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STUDY AND EVALUATION SCHEME
(With effective from academic session 2023-2024)
B.Tech. in Cloud Computing
YEAR III, SEMESTER VI

Sl. No.	Category	Course Code	Course Title/ Subjects	Hours per week			Evaluation Scheme		Total	Credits
				L	T	P	CA	EE		
THEORY										
1	Professional Core	BCSI601	Artificial Intelligence	3	0	0	25	50	75	3
2	Professional Core	BCSICT601	Linux Administration	1	0	0	10	15	25	1
3	Professional Core	BCSICT602	Ethical Hacking	3	1	0	30	70	100	4
4	Professional Core		Elective -II	3	1	0	30	70	100	4
5	Professional Core	BCSICT608	Incident Response Management	2	0	0	15	35	50	2
6	Open Elective		Open Elective-I	3	0	0	25	50	75	3
7	Engineering Science Course	IIOT6	Advanced Artificial Intelligence	4	0	0	30	70	100	4
PRACTICALS AND PROJECTS										
7	Professional Core	BCSICT612	Exploring Software as a Service (SaaS) Lab	0	0	2	20	30	50	2
8	Professional Core	SICT615	Linux administration Lab	0	0	2	20	30	50	2
9	Professional Core	SICT616	Ethical Hacking Lab	0	0	2	10	15	25	1
10	Project	BCSICT617	Project-I			4				
			TOTAL	19	2	12	215	435	650	26

L-Lecture, **T**- Tutorial , **P**- Practical ,**CA**- Continuous Assessment, **EE**- End Semester Examination

Course code	Open Elective - I
BCSICT609	UI/UX Fundamentals
SICT610	Mobile Application Development
BCSICT611	Business Intelligence

BCSICT 502 - Network Security

L T P C
3 0 0 3

MODULE-I

Computer Network System Security Introduction: Introduction, What is computer Network security, Sample Attacks Computer Network, The Marketplace for vulnerabilities, Error 404 Hacking, Security Policies and Security Handshake Pitfalls, What is security policy, high and low level policy, Protocol problems, assumptions, shared secret protocols, public key protocols, mutual authentication, reflection attacks, use of timestamps, nonce and sequence numbers, session Keys, one-and two-way public key based authentication, Authentication of People: Verification techniques, passwords, length of passwords, password distribution, smart cards, and biometrics

MODULE-II

IP Security: IP Security Overview, IP Security Policy, Encapsulating Security Payload, Combining Security Associations, Internet Key Exchange (IKE).Transport-Level Security: Web Security Considerations, Secure Sockets Layer, Transport Layer Security, HTTPS standard, Secure Shell (SSH) application. Malicious Software: Viruses, Worms, System Corruption, Attack Agents, Information Theft Key loggers, Phishing, Spyware Payload Stealthing, Backdoors, Rootkits, Distributed Denial of Service Attacks, Major web server threats ,Cross site request forgery ,Cross site scripting ,Defenses and protections against XSS , Finding vulnerabilities ,Secure development.

MODULE-III

Basic cryptography: Public key cryptography ,RSA public key crypto ,Digital signature Hash functions ,Public key distribution ,Real world protocols ,Basic terminologies ,Email security certificates ,Transport Layer security TLS ,IP security , DNS security. Basic security problems, Routing security, DNS revisited, Summary of weaknesses of internet security, .Link layer connectivity and TCP IP connectivity, Packet filtering firewall, Intrusion detection.

Text books:

1. William Stallings, Network Security Essentials: Applications and Standards, Prentice Hall, 4th edition, 2010.
2. Michael T. Goodrich and Roberto Tamassia, Introduction to Computer Security, Addison Wesley, 2011.
3. William Stallings, Network Security Essentials: Applications and Standards, Prentice Hall, 4th edition, 2010.
4. Alfred J. Menezes, Paul C. van Oorschot and Scott A. Vanstone, Handbook of Applied Cryptography, CRC Press, 2001

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BCSICT504 – Humanities II Engineering & Managerial Economics

MODULE-I

Introduction: Meaning, Nature and Scope of Economics, Meaning of Science, Engineering and Technology. Managerial Economics and its scope in engineering perspective.

MODULE-II

Basic Concepts Demand Analysis, Law of Demand, Determinates of Demand, Elasticity of Demand-Price, Income and cross Elasticity. Uses of concept of elasticity of demand in managerial decision

MODULE-III

Demand forecasting Meaning, significance and methods of demand forecasting, production function, Laws of returns to scale & Law of Diminishing returns scale. An overview of Short and Long run cost curves – fixed cost, variable cost, average cost, marginal cost, Opportunity cost

MODULE-IV

Market Structure Perfect Competition, Imperfect competition – Monopolistic, Oligopoly, duopoly sorbent features of price determination and various market conditions.

MODULE-V

National Income, Inflation and Business Cycles Concept of N.I. and Measurement. Meaning of Inflation, Type causes & prevention methods, Phases of business cycle

Reference Books:-

1. Koutsoyiannis A: Modern Microeconomics, ELBS.
2. Managerial Economics for Engineering: Prof. D.N. Kakkar
3. Managerial Economics: D.N. Dwivedi
4: Managerial Economics: Maheshwari.

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BCSICT608 Incident Response Management(Cyber Security)

MODULE-I

Need for CSIRM Differences between an event, incident and disaster, what are cyber security incidents, need for CSIRM, policy, plan and procedure, importance of communication protocol, key internal and external stakeholders, law enforcement, role of media, team structure and roles – important considerations.

MODULE-II

Handling a Cyber Security Incident Incident response lifecycle, incident handling infrastructure and facilities requirements, detection and analysis, process, tools and techniques, attack vectors, recognizing signs of an incident, precursors, indicators and historical organization data, incident correlation, review of logs and vital system parameters, incident handling checklist, documentation and reporting

MODULE-III

Recovering from Cyber Security Incidents Nature of incidents and the type of resources it affects, assessment of an incident's impact on business, IT operations and information, determining the amount of time and resources needed in recovering from an incident, prioritization, incident notification structure, containment, eradication and recovery – choosing a containment strategy, evidence gathering and handling, identifying the attack hosts, eradication and recovery, post-incident analysis, evidence retention and lessons learned

MODULE-IV

Preventing Cyber Security Incidents Incident analytics as input to proactive security measures to prevent incidents, risk assessment, host security, network security, malware prevention, user awareness and training, analysis of cost of control versus cost of incident impact, best practices.

MODULE-V

Cyber Security Incidents Analysis through Scenarios Flow chart of scenario questions, scenarios – DoS attack on DNS server, worm and DDoS agent infestation, military-classified documents stolen by an insider, compromised database server, unauthorized access to payroll records, identities and credentials stolen by hackers, antisocial propaganda in media through compromised home wifi network, personal files stored in Cloud are compromised, remote hacking of smart home network, malware infection in home and office network simultaneously, large scale of citizens' biometric data stolen by cyber war groups

Reference Books:-

Reference Books:

1. NIST SP 800-61r2 – Computer Security Incident Handling Guide
2. Computer Incident Response and Product Security (Networking Technology: Security) by DamirRajnovic 1st, Kindle Edition
3. Intelligence-Driven Incident Response: Outwitting the Adversary 1st Kindle Edition
4. The Computer Incident Response Planning Handbook: Executable Plans for Protecting Information at Risk by N.K. Mccarthy, Matthew Todd, Jeff Klaben, McGraw-Hill Education, 2012
5. Tools and Techniques for Fighting Malicious Code: Malware Analyst's Cookbook by Michael Hale Ligh, Steven Adair, Blake Hartstein, Matthew Richard, Wiley, 2010
6. Incident Response: A Strategic Guide to Handling System and Network Security Breaches by E. Eugene Schultz, Russell Shumway, Sams, 2001
7. The Effective Incident Response Team by Julie Lucas, Brian Moeller, Addison Wesley, 2003
8. Information Security: Incident Response and Disaster Recovery by Michael E. Whitman, Herbert Mattford,

Cengage Learning India Pvt Ltd, 2009
9. Crafting the InfoSec Playbook: Security Monitoring and Incident Response Master Plan 1st Kindle Edition by Jeff Bollinger, Brandon Enright, Matthew Valites, 2015
10. Incident Management and Response Guide: Tools, Techniques, Planning, and Templates Kindle Edition by Tom Olzak, Erudio Security, 2017
11. Cyber Security by Nina Godbole, SunitBelapure, Wiley, 2011
12. Incident Response & Computer Forensics by Jason T. Luttgens, Matthew Pepe, Kevin Mandia, McGraw-Hill Education; 3rd edition, 2014
13. Principles of Incident Response and Disaster Recovery by Michael Whitman, Herbert MattordDelmar Cengage Learning; 2nd Revised edition, 2013
14. Computer Incident Response and Forensics Team Management: Conducting a Successful Incident Response by Leighton Johnson, Syngress, 2013
15. Cyber Incident Response: Bridging the Gap Between Cybersecurity and Emergency Management by Response, and Communications and the Subcommittee on Cybersecurity, Infrastructure Protection, and Security Technologies of the Committee on Homeland Security House of Representatives Subcommittee on Emergency Preparedness, CreateSpace Independent Publishing Platform (8 May 2014)

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Open Elective-I

BCSICT609 UI/UX Fundamentals

Course Description: The increasing possibilities with interactive technology as opened to virtual classrooms for teaching and educating the students. Research has proven that interactive teaching using such visual technologies is much more effective than the traditional methods which help students understand and gain knowledge better. Virtual reality is used in many training scenarios as it consists of a wide range of benefits for academia and industrial needs. Course Objectives: Students will get to know about various techniques of Graphic Design and UI/UX and will develop skills to become a professional designer. They will be taught to enhance their knowledge and master tools producing good industry standard designs. Students will be able to work on advertisements, website, and app designs.

Course Outcomes (COs): At the end of this course students will be able to:
CO 1. Create Graphic Design artworks of your own.
CO 2. Explain the functionality of different design related software
CO 3. Use learned skills to solve problems of various layouts
CO 4. Test own's skill and knowledge for a better workflow
CO 5. Select best output and what works for a particular given project
CO 6. Develop ideas and various app designs and website pages..

MODULE-I

Unit 1: Computer Fundamentals & Digital Illustration, Introduction to Graphic Design and Its Uses, Raster & Vector Graphics, Drawing Vector Shapes and Illustrations, Art & Sketching, Drawing Techniques, Conceptual Thinking in Creativity, Developing a Personal Illustration Style, Color Modes, Schemes, Design, Image Retouching and Color Balancing, Using Filters Corporate Identity Design Designing Brochures & Catalogues Layouts for Newspapers, Designing Magazines, Visual Design Principles

MODULE-II

UI design fundamentals, Evolution of user interfaces, Interaction with physical components, Flat design, Role of UI in UX,Laws of digital interface design, Understand user experience,VIMM model, Know your user, user research, Difference between design and art, emotional design, designing for mental models, Importance of presentation, content, interactions, screen elements, accessibility, and global standards

MODULE-III

Typography, Types of typefaces, Typography Terminology, Guidelines for proper type selection, Typography design, Analyzing Aesthetics as per laws of Design principles, Alignment, Spacing, Lighting & Shadows, Grids,Consistency

MODULE-IV

Design Language & Rapid Prototyping • In-depth study of Design languages for different Google material platforms • Style guides and its importance • Apple Human centered Design guidelines measurements of UI components • Design for platforms: Mobile, Web, Tablet, Responsive, Smart Watch • Mood boards • UX design principles and laws • Introduction to Adobe XD • Creating low and high-fidelity prototypes

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BCSICT610 Mobile ApplicationDevelopment

MODULE-I

Introduction to Android: The Android Platform, Android SDK, Eclipse Installation, Android Installation, Building you First Android application, Understanding Anatomy of Android Application, Android Manifest file..

MODULE-II

Android Application Design Essentials: Anatomy of an Android applications, Android terminologies, Application Context, Activities, Services, Intents, Receiving and Broadcasting Intents, Android Manifest File and its common settings, Using Intent Filter, Permissions.

MODULE-III

Android User Interface Design Essentials: User Interface Screen elements, Designing User Interfaces with Layouts, Drawing and Working with Animation.

MODULE-IV

Testing Android applications, Publishing Android application, Using Android preferences, Managing Application resources in a hierarchy, working with different types of resources. UNIT - V Using Common Android APIs: Using Android Data and Storage APIs, Managing data using Sqlite, Sharing Data between Applications with Content Providers, Using Android Networking APIs, Using Android Web APIs, Using Android Telephony APIs, Deploying Android Application to the World..

TEXT BOOKS: 1.

T1. Lauren Darcey and Shane Conder, “Android Wireless Application Development”, Pearson Education, 2nd ed. (2011)

REFERENCE BOOKS:

R1. Reto Meier, “Professional Android 2 Application Development”, Wiley India Pvt Ltd
R2. Mark L Murphy, “Beginning Android”, Wiley India Pvt Ltd
R3. Android Application Development All in one for Dummies by Barry Burd, Edition: I

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BCSICT616 PC Ethical Hacking Lab

Course Objectives:

Introduces the concepts of Ethical Hacking

Gives the students the opportunity to learn about different tools and techniques in Ethical hacking and security Practically apply Ethical hacking tools to perform various activities.

Course Outcomes:

After completion of course, students would be able to:

Understand the core concepts related to vulnerabilities and their causes

Understand ethics behind hacking and vulnerability disclosure

Appreciate the impact of hacking

Perform the lab Experiments based on the following facts.(Develop your own set of lab experiments)

1 Google Hacking

2 Scanning

- Locating Open Ports
- Network mapping
- OS Fingerprinting

3 Gaining and Maintaining Access
4 Passwords hacking
5 Password Cracking Methods
6 Password Cracking Software
7 Man-in-the-Middle
8 Backdoors
9 Denial of Service
10 Covering tracks
11 Intrusion Detection Systems
12 Intrusion Prevention Systems
13 Anti-viruses
14 Malware
15 Viruses